EDUNET AICTE INTERNSHIP

AI AGENT FOR CHRONIC DISEASE MONITORING

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OUTLINE

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PROBLEM STATEMENT

An AI agent for chronic disease monitoring helps patients and healthcare providers manage long-term conditions effectively. It continuously analyzes health data from wearables, medical records, and patient inputs to detect early warning signs. Using AI and predictive analytics, it offers personalized insights, medication reminders, and lifestyle recommendations. The agent supports diseases like diabetes, hypertension, and heart conditions with real-time monitoring and alerts. It enables proactive care, reduces hospital visits, and improves patient adherence to treatment plans. This intelligent assistant bridges the gap between patients and providers, enhancing chronic care outcomes.

Proposed Solution:

Develop an AI-powered agent that continuously monitors chronic diseases by analyzing data from wearables, medical records, and patient inputs. It provides real-time alerts, personalized health insights, medication reminders, and lifestyle recommendations for conditions like diabetes, hypertension, and heart diseases.

TECHNOLOGY USED

IBM cloud lite services

Natural Language Processing (NLP)

Vector Indexing

Artificial Intelligence

IBM Granite model

IBM CLOUD SERVICES USED

- IBM Cloud Watsonx AI Studio
- IBM Cloud Watsonx AI runtime
- IBM Cloud Agent Lab
- IBM Granite foundation model
- Cloud Object Storage

WOW FACTORS

- This agent will revolutionize chronic disease care by enabling real-time health tracking, reducing hospital admissions, improving treatment adherence, and empowering both patients and providers through intelligent, personalized insights.

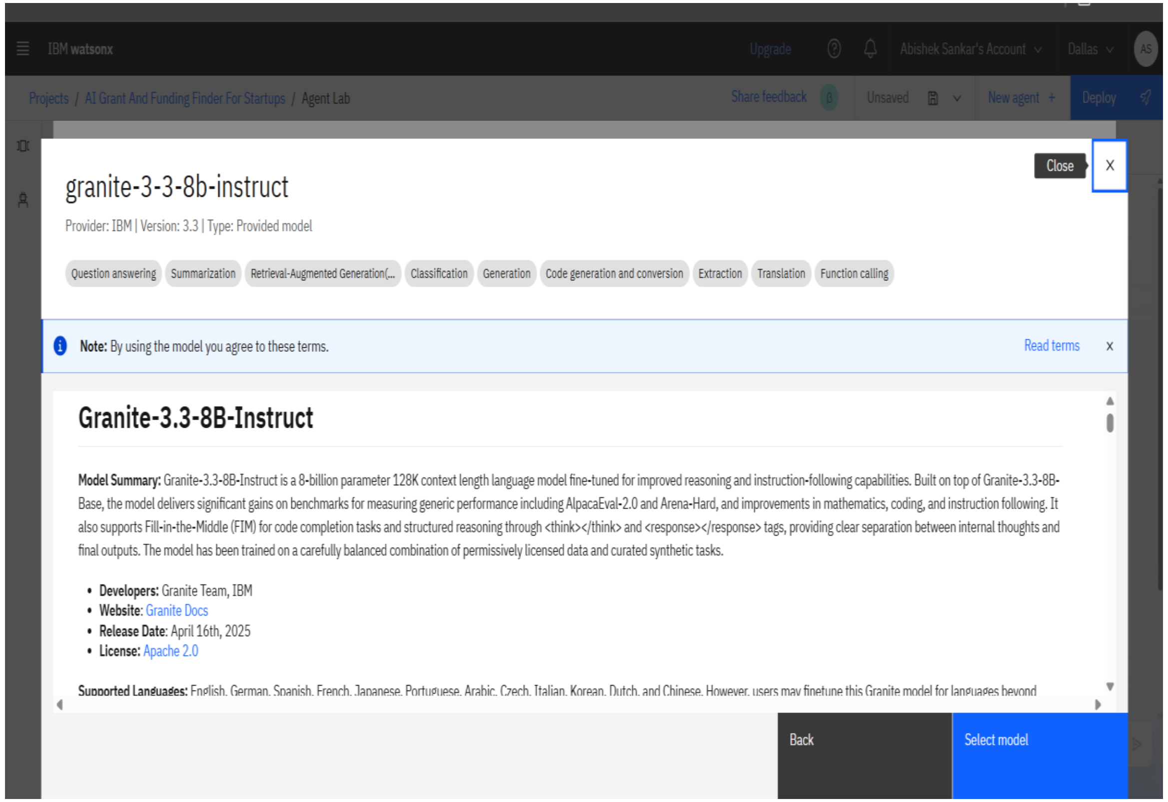
Unique Features:

- Continuous monitoring via wearable and health record integration
- Personalized recommendations for medication, diet, and lifestyle
- Real-time alerts for critical health deviations
- Predictive analysis for early detection of complications
- Daily symptom tracking with natural language input
- Smart reminders for medication and health check-ins
- Secure data sharing with healthcare providers for better collaboration

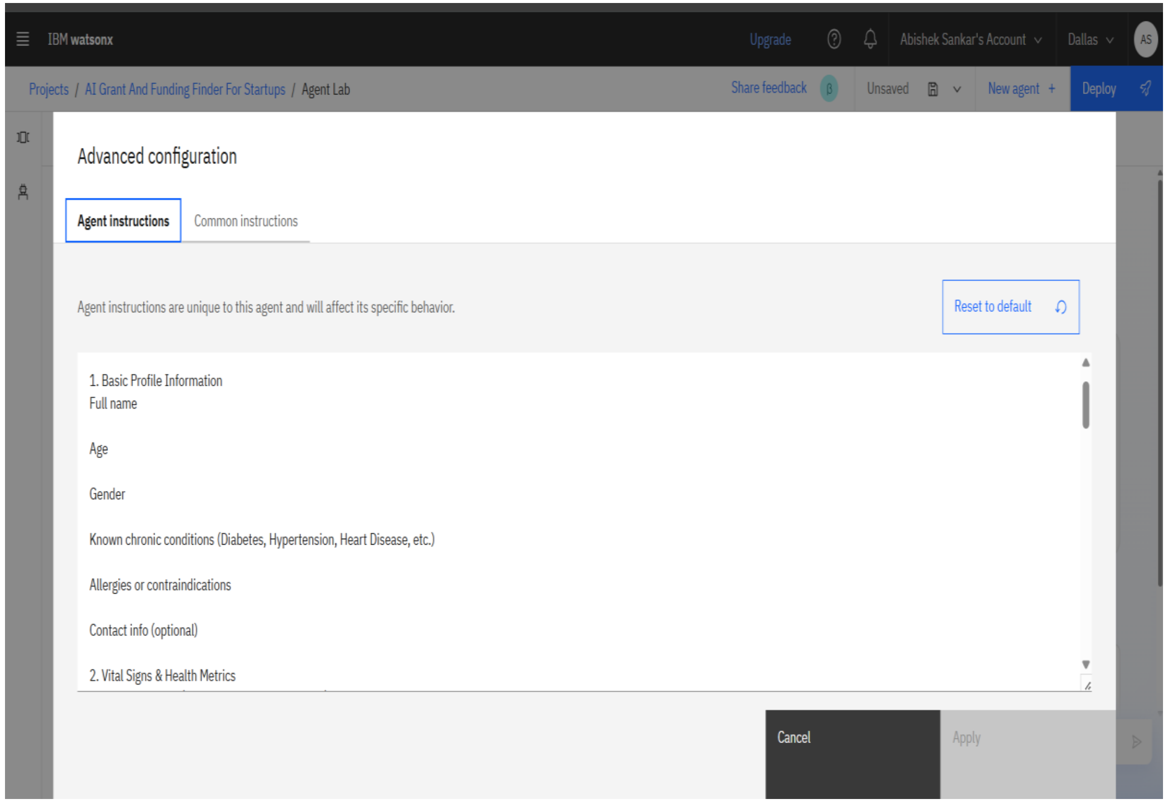
END USERS

- Patients with chronic conditions (e.g., diabetes, hypertension, heart disease)
- Caregivers and family members
- Healthcare Providers(Doctors , Nurses, Specialists)
- Hospitals and Clinics
- Telemedicine platforms
- Health insurance companies
- Government and public health agencies

RESULTS



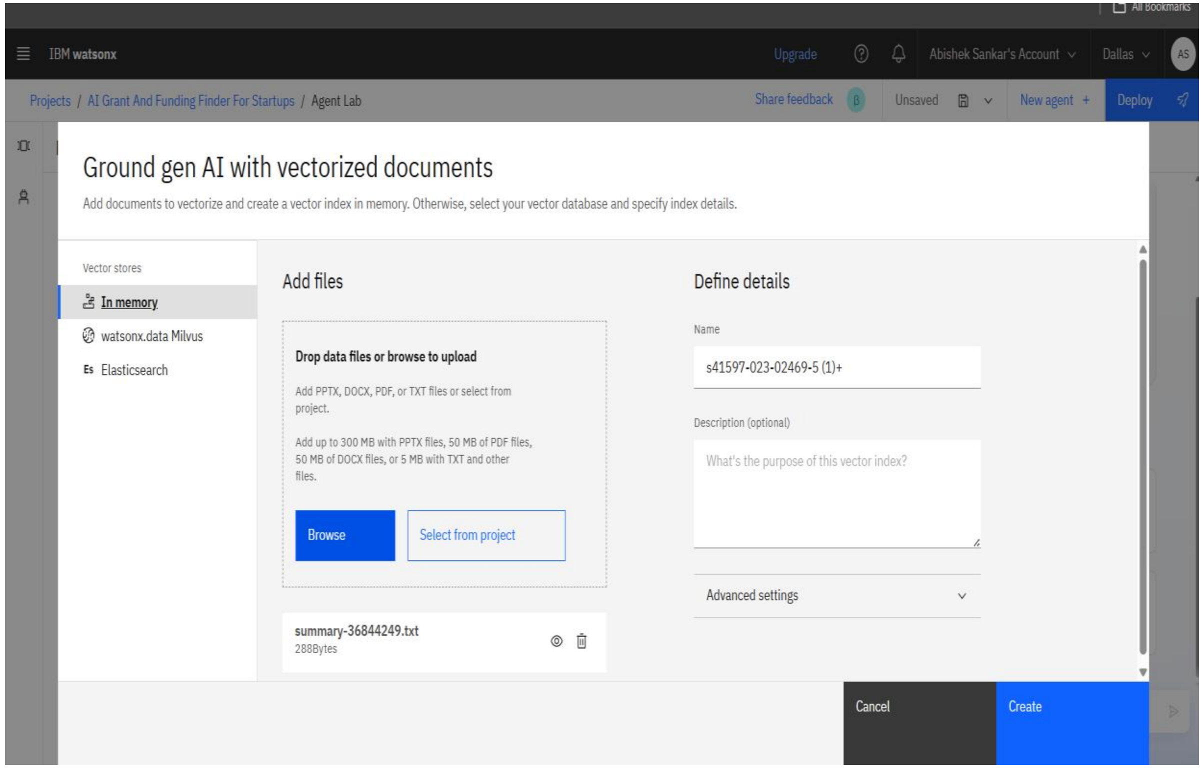
Using Granite Model



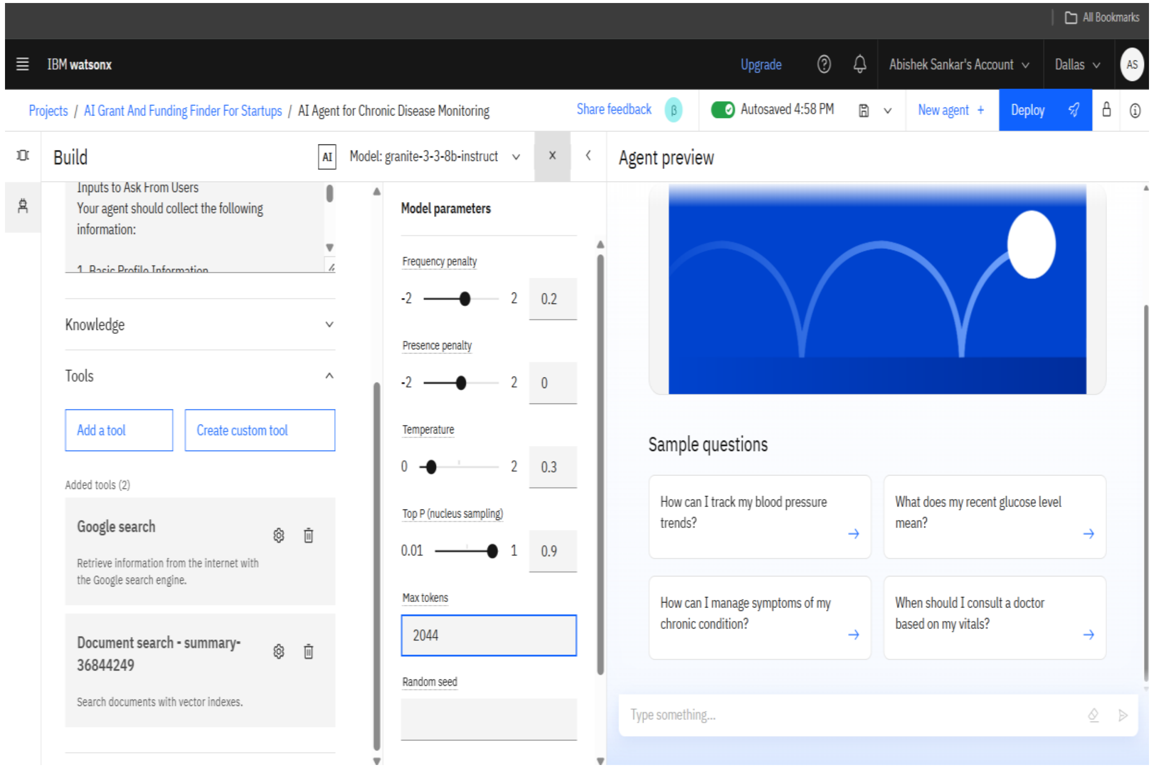
Adding Instructions

edunet foundation

RESULTS

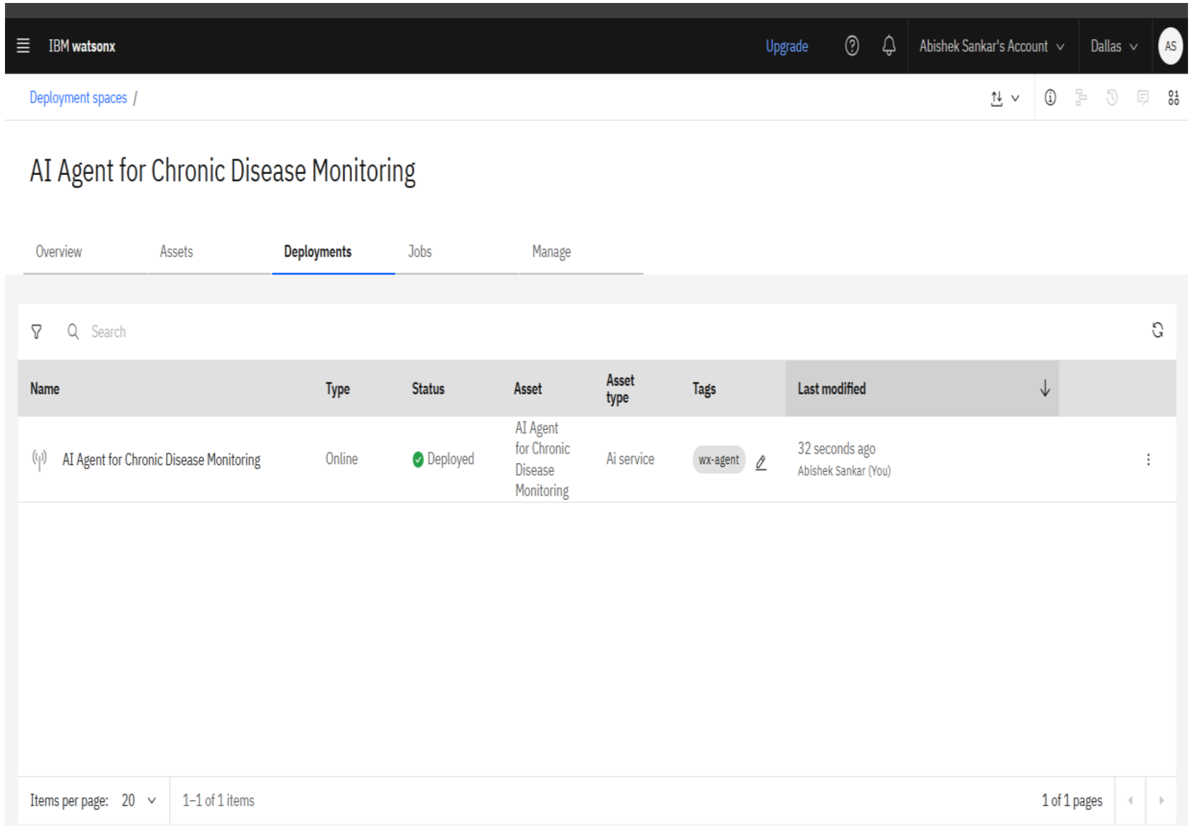


Vectorized Documents

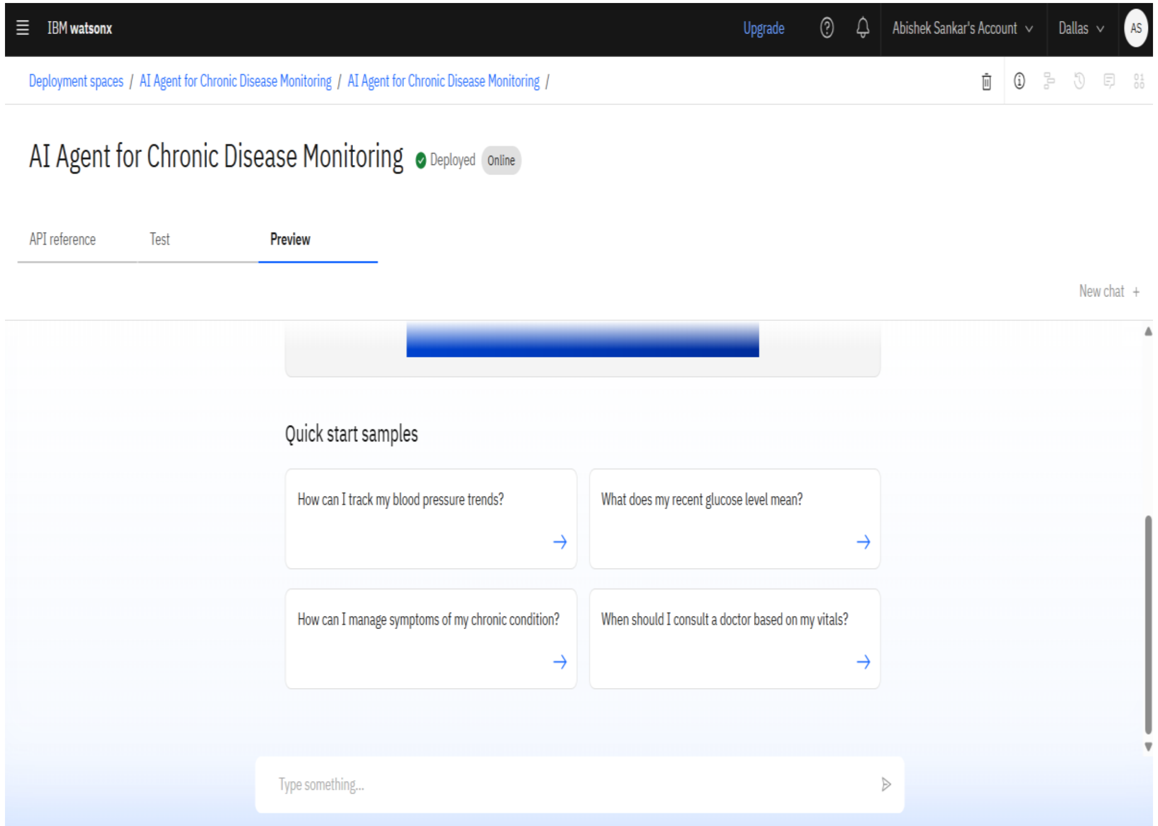


Model Parameters

RESULTS

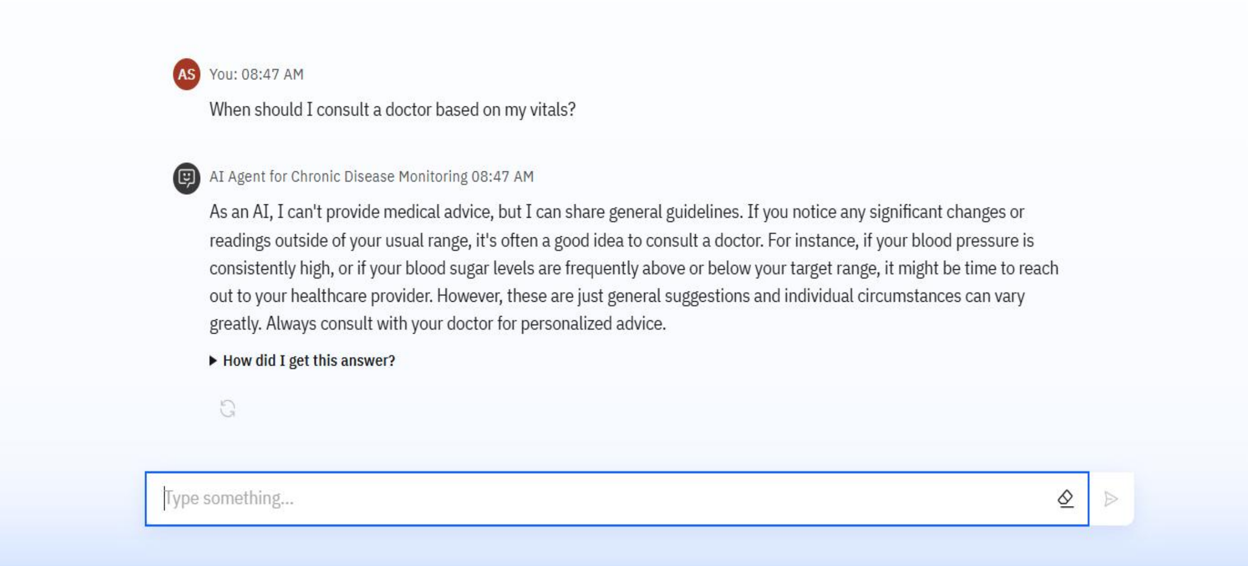


Agent Deployed

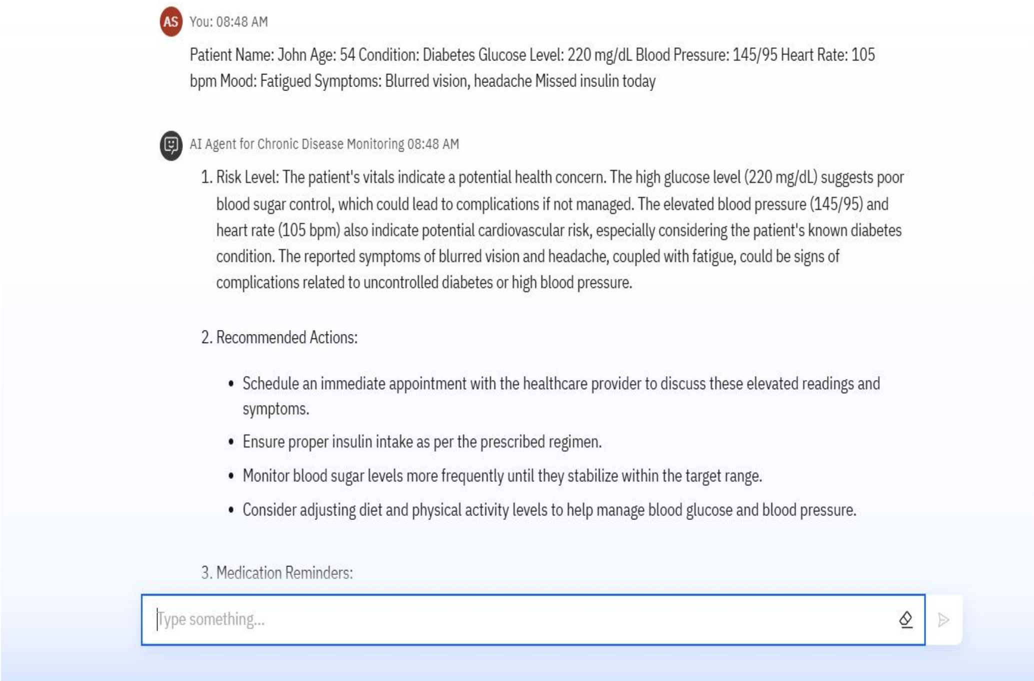


Sample Questions

RESULTS



Answer Generated For Sample Question



Customized User Input and Generated Result

CONCLUSION

- The agent delivers personalized insights, monitors patient vitals in real-time, and provides timely alerts to prevent complications.
- It enhances chronic care by automating follow-ups, managing health data, and supporting informed clinical decisions.
- Health AI agents boost efficiency, adherence, and proactive care across both individual and institutional healthcare systems.

GITHUB LINK

LINK : https://github.com/ABISHEK-S-15/edunet-intern.git

FUTURE SCOPE

• Multilingual Health Support for regional and global accessibility
• Voice – Enabled Health Interaction for elderly and visually impaired patients
• Real-Time Caregiver Collaboration and shared dashboards
• Predictive Risk Modeling for emerging comorbidities
•Integration with Telemedicine and HER Systems
• AI-Assisted Report Generation for clinicians and insurance claims

IBM CERTIFICATIONS

Getting Started with Artificial Intelligence

IBM SkillsBuild

IBM

In recognition of the commitment to achieve professional excellence

Abishek S

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Jul 16, 2025
Issued by: IBM SkillsBuild

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IBM **SkillsBuild** Completion Certificate

This certificate is presented to

ABISHEK S

for the completion of

Lab: Retrieval Augmented Generation with LangChain

(ALM-COURSE_3824998)

According to the Adobe Learning Manager system of record

Completion date: 16 Jul 2025 (GMT)

Learning hours: 20 mins

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THANK YOU